AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM256
Cyber Security

Midterm Examination (MTA)
Fall Semester 2025/2026

Date: 19.Nov./ 2025

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 2 Hours

Instructions:

- Total Marks: **30**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: Multiple Choice Questions [10 Marks]

Question 1: Choose the correct answer for the following statements:

1. Which of the following best defines a vulnerability?
   a) A potential event that could cause harm
   b) A weakness that a threat can exploit
   c) A method used to mitigate risk
   d) A type of patching process

2. Which of these is an example of a threat actor?
   a) Encryption algorithm
   b) Malware
   c) Hacker
   d) Firewall

3. What is risk in the context of cybersecurity?
   a) The act of patching systems
   b) The likelihood that a vulnerability will be exploited
   c) The process of encrypting data
   d) A security policy

4. Which of the following is a mitigation strategy for known vulnerabilities?
   a) Ignoring software updates
   b) Disabling encryption
   c) Applying security patches promptly
   d) Allowing default passwords

5. Which term describes the period between the discovery of a vulnerability and its patching?
   A. Zero-day
   B. Attack surface
   C. Window of vulnerability
   D. Cyber Kill Chain

6. Which principle ensures that data cannot be modified without detection?
   a) Integrity
   b) Confidentiality
   c) Availability
   d) Non-repudiation

7. Asymmetric cryptography relies on:
   a) A shared secret key
   b) Two mathematically related keys
   c) Random numbers only
   d) A single private key

8. Which of the following is a public-key algorithm?
   a) AES
   b) DES
   c) RSA
   d) SHA-256

9. Authorization determines:
   a) Who the user is
   b) What actions the user is allowed to perform
   c) How data is encrypted
   d) The strength of a password

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10. Which of the following is a common operating system hardening practice?
    a) Installing unverified software
    b) Disabling automatic updates
    c) Removing unnecessary services and accounts
    d) Using the same password for all accounts

PART 2: Open-Ended Questions [11 Marks]

Question 2: Discuss the importance of the CIA triad in designing a secure information system. [1.5 Marks]

Question 3: Why are humans often referred to as the "weakest link" in cybersecurity, and how can training reduce this risk? [2 Marks]

Question 4: Discuss the limitations of symmetric encryption systems. [1.5 Marks]

Question 5: How does a digital certificate support trust on the Internet? [2 Marks]

Question 6: Explain the concept of hashing and why it is essential for password storage. [2 Marks]

Question 7: Explain the principle of least privilege and its importance in operating-system security. [2 Marks]

PART 3: Problems [9 Marks]

Question 8: A company developing a secure messaging app decides to encrypt messages using AES with a fixed, hard-coded key stored in the source code. To facilitate communication between users, messages are also signed with the same key. The app becomes popular but soon suffers from unauthorized message interception and spoofing. **[4 Marks]**

1. Evaluate the cryptographic design flaws in this implementation.
2. Recommend a secure alternative architecture for confidentiality and authentication.
3. Explain how proper key management could have prevented the compromise.

Question 9: During an online shopping session, a browser warning appears: *"This website's SSL certificate is self-signed and not trusted."*. The user ignores it and completes the payment. Days later, fraudulent charges appear on their account. **[5 Marks]**

1. Analyze what could have happened from both a technical and behavioral perspective.
2. Explain the role of digital certificates and Certificate Authorities in preventing such attacks.
3. Recommend end-user and organizational measures to avoid repetition.

End of questions

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